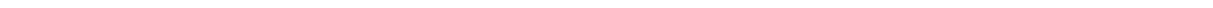


Task 9: Vulnerability Assessment Report

Application: Vuln Bank

Date: October 29, 2025

Submitted by: R S Abhinav



1 Introduction

This report details the findings of a brief security assessment conducted on the Vuln Bank web application. The application was deployed locally using Docker for educational purposes. The objective was to identify potential security vulnerabilities within a limited timeframe.

2 Methodology

The assessment involved the following steps:

1. Cloning the application repository from GitHub (<https://github.com/Commando-X/vuln-bank.git>).
2. Deploying the application using Docker and Docker Compose with the command: `docker-compose up --build -d`.
3. Accessing the running application via a web browser at `http://localhost`.
4. Performing manual inspection and testing for common web vulnerabilities, focusing on input validation and authentication mechanisms.

3 Setup Evidence

[illegible]

Figure 1: Output from `docker-compose up` command during setup.

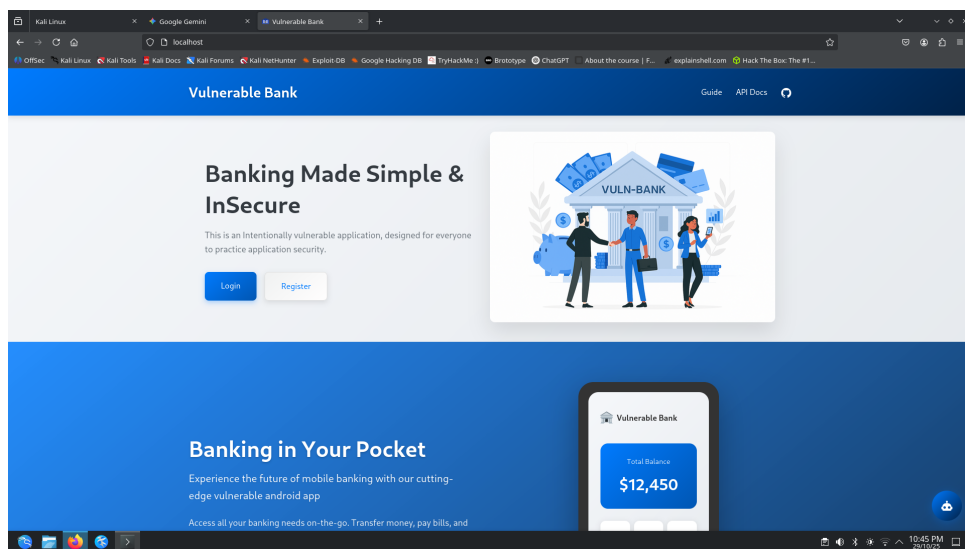


Figure 2: Initial Vuln Bank login page accessed at `http://localhost`.

4 Findings

The following vulnerability was identified during the assessment:

4.1 SQL Injection - Authentication Bypass

- **Location:** Login Page (/login)
- **Description:** The login functionality does not properly sanitize user input for username and password fields. This allows an attacker to manipulate the backend SQL query to bypass the authentication check.
- **Reproduction Steps:**
 1. Navigate to the login page (<http://localhost>).
 2. Enter the payload ' OR '1'='1 into the Username field.
 3. Enter the same payload ' OR '1'='1 into the Password field.
 4. Click the Login button.
 5. Observe successful login without valid credentials.
- **Risk: Critical.** This vulnerability allows any attacker to gain unauthorized access to potentially any user's account, compromising data confidentiality and integrity.
- **Recommendation:** Implement parameterized queries (prepared statements) for all database interactions involving user input. Input validation should be used as a secondary defense to reject unexpected characters or formats.
- **Evidence:**

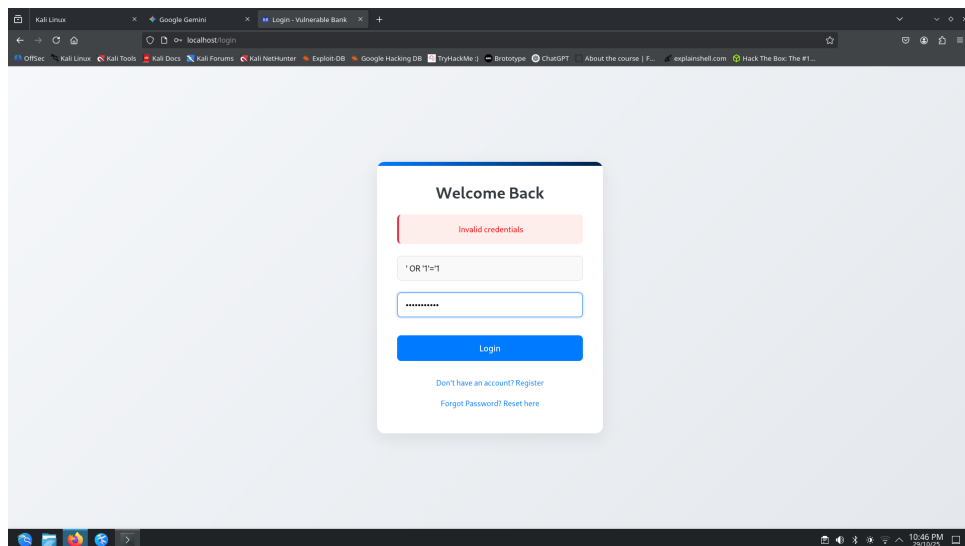


Figure 3: Login page showing the SQL Injection payload entered.

5 Conclusion

The Vuln Bank application, in its current state, contains at least one critical vulnerability (SQL Injection) that allows trivial bypass of authentication. This highlights the importance of secure coding practices, particularly proper input validation and the use of parameterized queries, to prevent common web application attacks. Further testing is likely to reveal additional vulnerabilities.

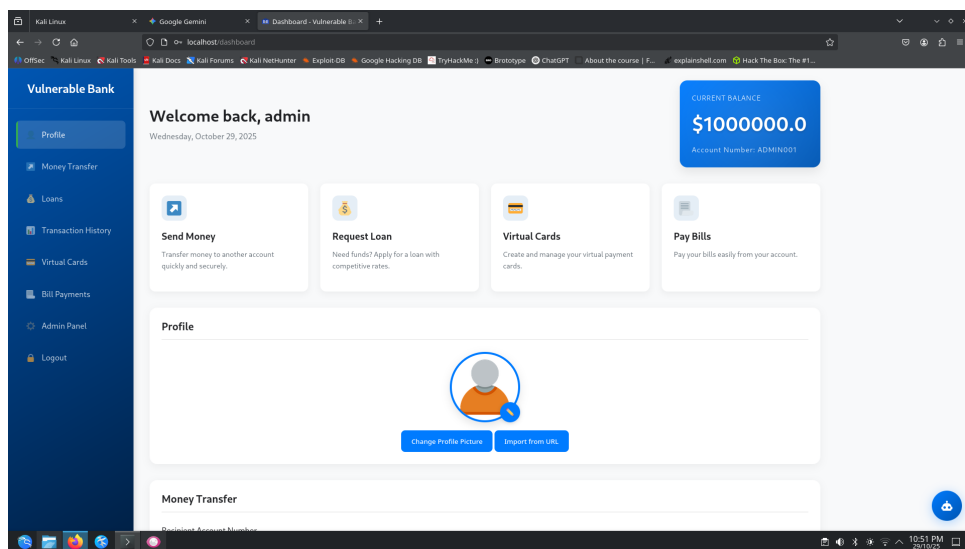


Figure 4: Successful login bypass showing the post-login page.